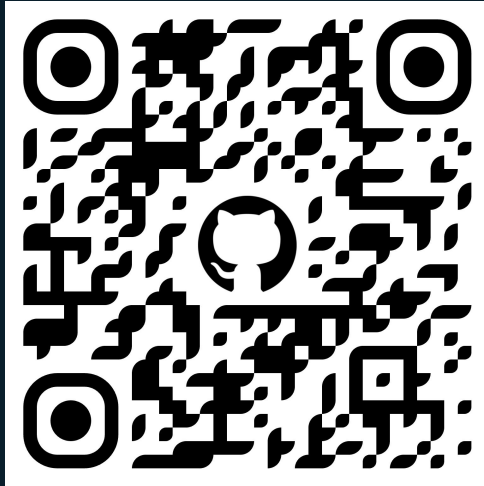


“Programming”

<Recap>

Resource



github.com/ro-witthawin/TBS-BI303-2026

TBS-BI303-2026



Class schedule

Date	Topic	Material
2026-08-03	Intro to Programming Language	Day01
2026-08-10	Basic String & List	Day02
2026-08-17	Function	Day03
2026-08-24	Loop	Day04
2026-08-31	Dictionary	Day05
2026-09-07	Class Object	Day06
2026-09-14	Freq Library	Day07
2026-09-21	Final Exam	

Grading

A : ≥ 80

B+ : 75-79

B : 70-74

C+ : 65-69

C : 60-64

D+ : 55-59

D : 50-54

F : < 50

[2026-08-31] Exam #1 : 40%

- String & List
- Function
- Loop

[2026-09-21] Exam #2 : 60%

- String & List
- Function
- Loop
- Dict
- Class

Logical Thinking

$$Q : 1 + 2 + (3 + 4) = ?$$

Programing Step

```
x = 15

if x > 10:
    print("x is greater than 10")
elif x == 10:
    print("x is exactly 10")
else:
    print("x is less than 10")
```

Basic Variable

Variable Naming rules

x = "I" 

X0 = "love" 

x_1 = "program" 

2x = "ing" ✗

x = True # Boolean

x = False # Boolean

x = 10 # Integer

x = 10.0 # Float

x = "Hello" # String

Basic Command & Operation

```
print(".....")  
print('.....')
```

```
type(...) # Check data type  
bool(...) # Change data to Boolean  
int(...) # Change data to Integer  
float(...) # Change data to Float
```

```
x = 10  
print(x)  
# Note & Comment  
print("Hello") # Note
```

Basic Operation

$x = 1 > 2$ # Greater than

$x = 2 < 3$ # Less than

$x = 3 == 4$ # Equal to

$x = 4.4 != 5$ # Not equal to

$x = 5 >= 6$ # Greater or equal

$x = 6 <= 7$ # Less or equal

$x = 1 + 2$ # Addition

$x = 2 - 3$ # Subtraction

$x = 3 * 4$ # Multiplication

$x = 4 / 5$ # Division

$x = 5 // 6$ # Floor Division

$x = 6 \% 7$ # Modulus

$x = 7 ** 8$ # Exponentiation

$x += 10$ # Short of $x = x + 10$

$x -= 10$ # Short of $x = x - 10$

$x *= 10$ # Short of $x = x * 10$

$x /= 10$ # Short of $x = x / 10$

And, Or, Not

x = True and True

x = False or True

x = not True

X = not(False)

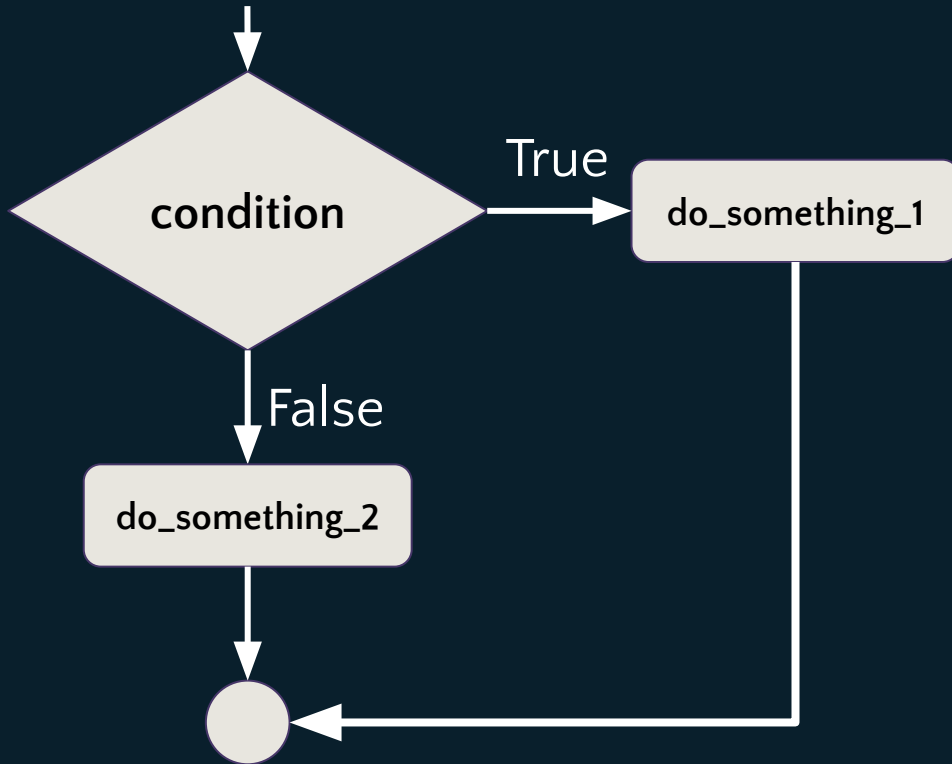
X	Y	X & Y
True	True	True
True	False	False
False	True	False
False	False	False

And

X	Y	X Y
True	True	True
True	False	True
False	True	True
False	False	False

Or

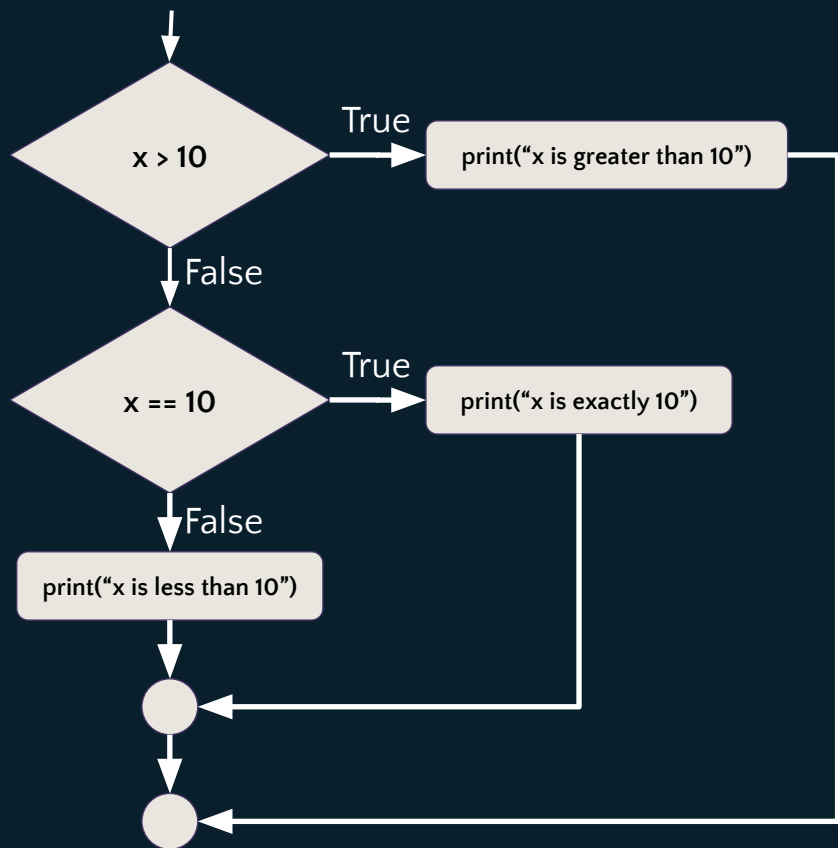
If / Else



```
x = 15
```

```
if x > 10:  
    print("x > 10")  
else:  
    print("x <= 10")
```

If / Else



```
x = 15
```

```
if x > 10:
    print("x is greater than 10")
else:
    if x == 10:
        print("x is exactly 10")
    else:
        print("x is less than 10")
```

```
x = 15
```

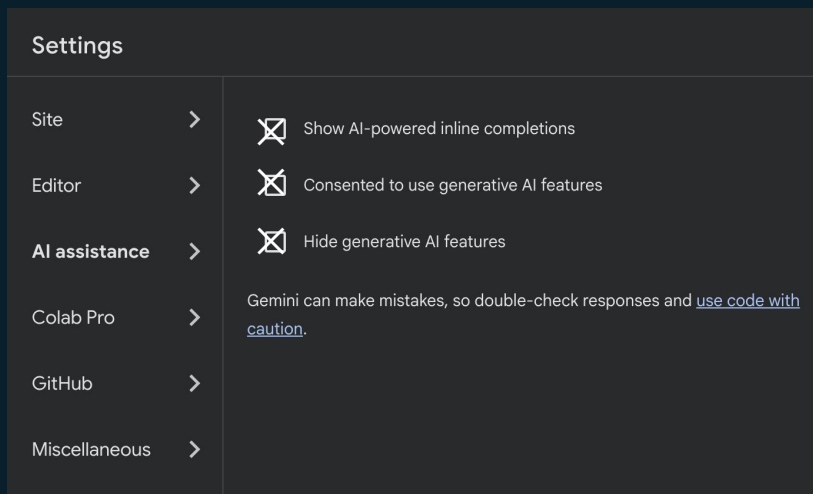
```
if x > 10:
    print("x is greater than 10")
elif x == 10:
    print("x is exactly 10")
else:
    print("x is less than 10")
```

Hand-on 15 min without AI



Credit Approval

- ถ้า Credit Score ≥ 700 และ รายได้ต่อเดือน $\geq 50,000$ ให้อนุมัติทันที
- ถ้า Credit Score < 600 หรือ รายได้ต่อเดือน $< 20,000$ ให้ปฏิเสธ
- นอกนั้นส่งให้เจ้าหน้าที่ตรวจสอบเพิ่มเติม



Input : credit_score, income

Output : status # ["approve", "reject",
"wait for review"]

“String”

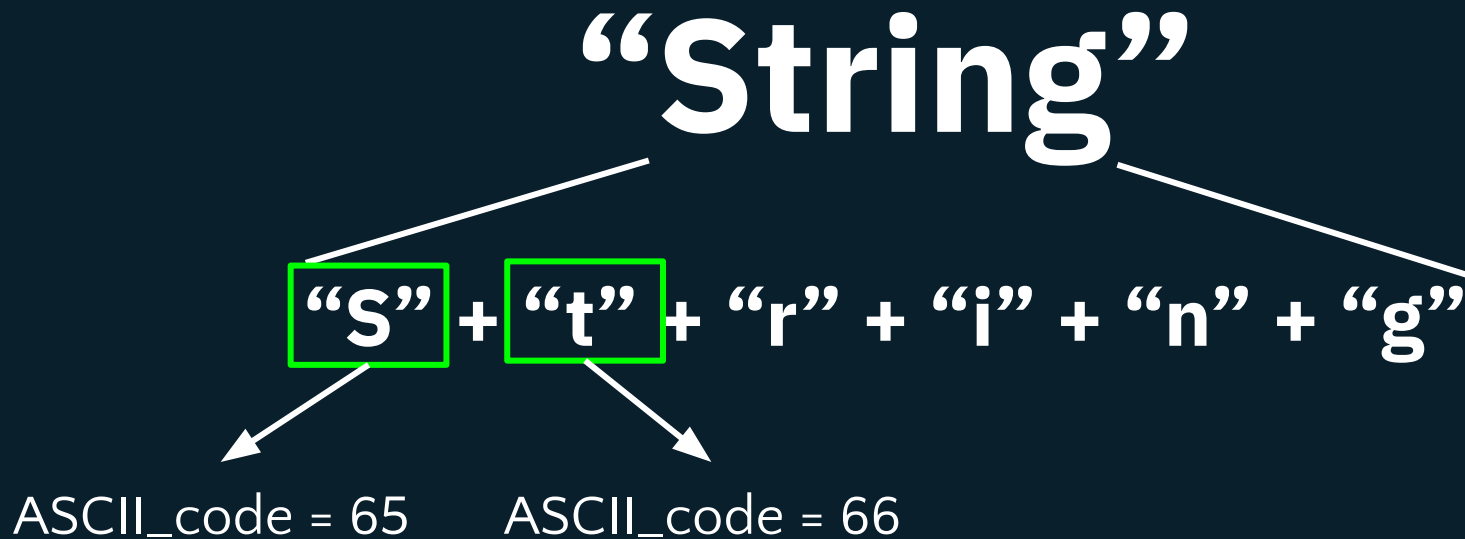
String

```
print(".....")  
print('.....')
```

“String”

“S” + “t” + “r” + “i” + “n” + “g”

String



String

```
print(".....")  
print('.....')
```

“String”

“S” + “t” + “r” + “i” + “n” + “g”

Index

[0]

[1]

[2]

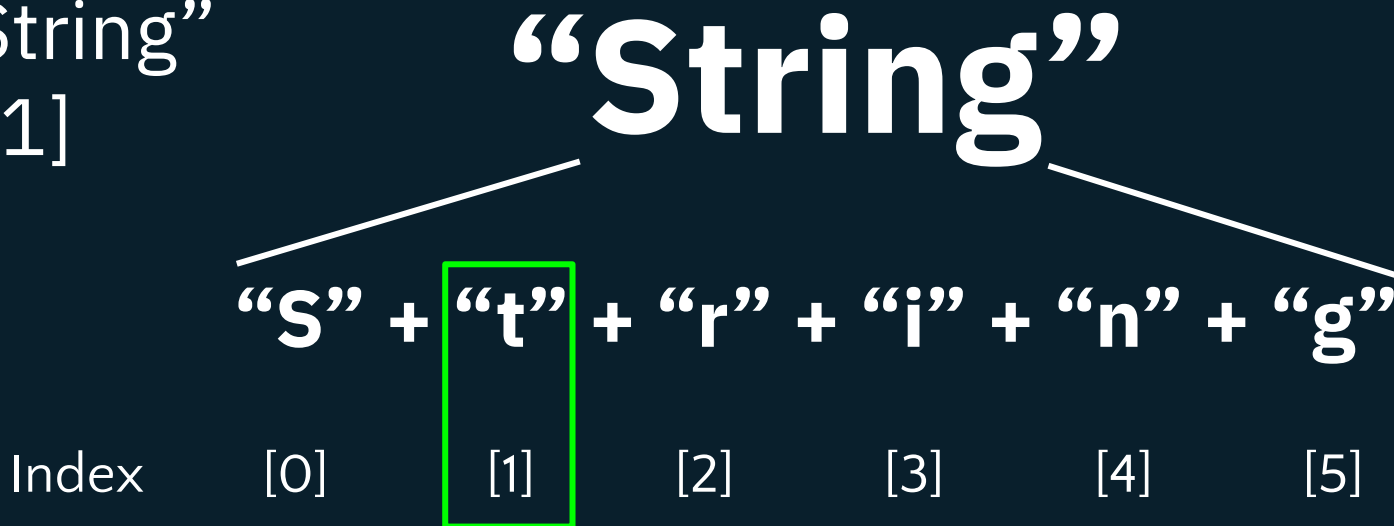
[3]

[4]

[5]

String

```
a = "String"  
b = a[1]
```



```
b = "t"
```

String

```
a = "String"  
b = a[0: 3]
```

"String"

"S" + "t" + "r" + "i" + "n" + "g"

Index

[0]

[1]

[2]

[3]

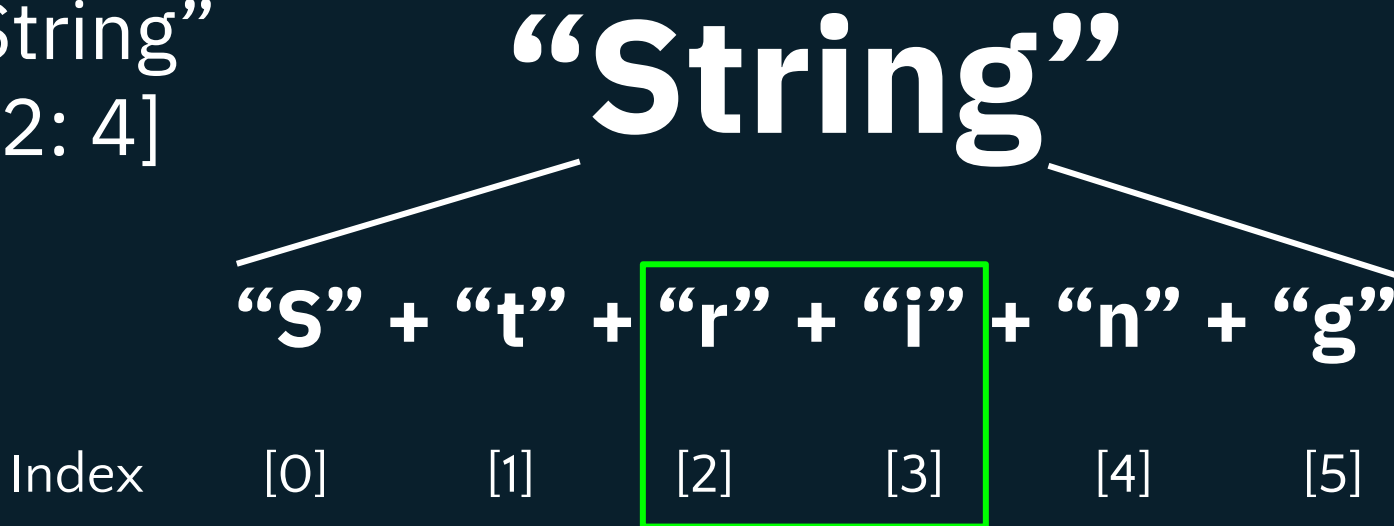
[4]

[5]

```
b = "Str"
```

String

```
a = "String"  
b = a[2: 4]
```



```
b = “ri”
```

String

```
a = "String"  
b = a[4:]
```

“String”

“S” + “t” + “r” + “i” + “n” + “g”

Index

[0]

[1]

[2]

[3]

[4]

[5]

```
b = “ng”
```

String

```
a = "String"  
b = a[:4]
```

"String"

"S" + "t" + "r" + "i" + "n" + "g"

Index

[0]

[1]

[2]

[3]

[4]

[5]

```
b = "Stri"
```


String

“String”

“S” + “t” + “r” + “i” + “n” + “g”

Index

[0]

[1]

[2]

[3]

[4]

[5]

Negative Index

[-6]

[-5]

[-4]

[-3]

[-2]

[-1]

String

“String”

a = “String”

b = a[-3:]

“s” + “t” + “r” + “i” + “n” + “g”

Index	[0]	[1]	[2]	[3]	[4]	[5]
-------	-----	-----	-----	-----	-----	-----

Negative Index	[-6]	[-5]	[-4]	[-3]	[-2]	[-1]
----------------	------	------	------	------	------	------

String: Indexing

“String”

a = “String”

b = a[-3:]

“s” + “t” + “r” + “i” + “n” + “g”

Index

[0]

[1]

[2]

[3]

[4]

[5]

Negative Index

[-6]

[-5]

[-4]

[-3]

[-2]

[-1]

b = “ing”

String: Indexing

“String”

a = “String”

b = a[-3:-2]

“s” + “t” + “r” + “i” + “n” + “g”

Index

[0]

[1]

[2]

[3]

[4]

[5]

Negative Index

[-6]

[-5]

[-4]

[-3]

[-2]

[-1]

b = “i”

String: Indexing

“String”

a = “String”

b = a[-3:-5]

“s” + “t” + “r” + “i” + “n” + “g”

Index	[0]	[1]	[2]	[3]	[4]	[5]
-------	-----	-----	-----	-----	-----	-----

Negative Index	[-6]	[-5]	[-4]	[-3]	[-2]	[-1]
----------------	------	------	------	------	------	------

b = “”

String: Operation

`"Hello Python".upper()` # "HELLO PYTHON"

`"Hello Python".lower()` # "hello python"

`" Hello Python ".strip()` # "Hello Python"

`"Hello Python".replace("Python", "World")` # Hello World

`"Hello Python".count("l")` # 2

`"Hello Python".find("o")` # 4

`str(3.14)` # "3.14"

“List”

List

[0, 1, 2, “a”, “B”]

List

[0, 1, 2, “a”, “B”]

Index	[0]	[1]	[2]	[3]	[4]
-------	-----	-----	-----	-----	-----

Negative Index	[-5]	[-4]	[-3]	[-2]	[-1]
-------------------	------	------	------	------	------